

AMIT VERMA

ML Engineer · Python Developer · Computer Vision · NLP · LLMs

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📍 Kolkata, India

PROFILE

Final-year CS undergraduate with DRDO (SSPL) research experience and a peer-reviewed publication at ICSSAI 2025. Specialises in LLM fine-tuning, RAG pipelines, transformer-based NLP, medical imaging AI, and computer vision. Built local ML systems with memory support, leveraging mixed-precision training, FAISS/ChromaDB vector retrieval, and local LLM deployment (Llama 3.1, Qwen 2.5) via llama-cpp on hardware.

TECHNICAL SKILLS

Languages

Python (Advanced) · C++ · JavaScript

ML / Deep Learning

PyTorch · Transformers · scikit-learn · Multimodal Models

Computer Vision

OpenCV · GradCAM / GradCAM++ · Thermal Imaging · Landmark Detection

Python Ecosystem

Flask · FastAPI · NumPy · Pandas · Matplotlib · PyQt

Tools & Infra

Git · Linux · REST APIs

EDUCATION

B.Tech in Computer Science and Business Systems

Techno India University, Kolkata Aug 2022 – Apr 2026

CGPA: 8.1 / 10

LANGUAGES

English ▪ Hindi ▪ Bengali

RESEARCH & ACHIEVEMENTS

Publication — ICSSAI 2025

Comparative study of image denoising under Gaussian noise; Group Normalization achieved **7.9% higher PSNR** vs. Batch Normalization. Accepted & presented at *ICSSAI 2025*.

Amazon ML Challenge

End-to-end **multimodal pipeline** for product attribute prediction on **2M+ samples**: MiniLM text encoding and EfficientNet-B0 image features fused into a shared embedding space; deep MLP head with BatchNorm & Dropout for robust regression to predict price from description and images.

Rank 1570 · Score 51.20 · [GitHub](#)

EXPERIENCE

Software Engineering Intern

Feb -- May 2025

TIU – DRDO Executive Lab

- Built an **image processing application** with **50+ filters** and integrated **N2N, N2V, DnCNN**, improving **PSNR/SSIM by 10%**.
- Developed a **multithreaded video pipeline** with real-time controls, reducing latency by **35%**.
- Processed **5D thermal (.mat) data** using **NUC** to generate smooth thermal videos.
- Fixed crashes and memory leaks, improving system stability.

PROJECTS

1. Local LLM Memory System with RAG

[GitHub](#)

Llama 3.1 8B · Qwen 2.5 14B · llama-cpp · ChromaDB · sentence-transformers · Python

- Deployed Llama 3.1 8B and Qwen 2.5 (14B/7B) locally via llama-cpp; optimized inference through layer-wise offloading, and wrapped the UI using **Tauri** for a lightweight desktop interface with zero API cost inference.
- Built persistent memory architecture using **ChromaDB** + **sentence-transformers** for semantic retrieval across sessions.
- Designed modular RAG pipeline with context-aware retrieval for long-horizon conversations and document

2. Multitask Sentiment Analysis with Sarcasm Detection

[GitHub](#)

XLNet-RoBERTa · HuggingFace · Multitask Learning · FAISS · Mixed Precision

- 93% validation accuracy** on **200k+ samples** (Reddit, iSarcasm, humor) with sarcasm-aware gating mechanism.
- 3 polarity FAISS sub-indexes** using CLS + mean-pool vectors; eliminated cross-polarity noise in emotion matching.
- Resolved NaN instability via adaptive gradient clipping and inverse-frequency class weighting; effective batch **64**.

3. Brain Tumor Classification with GradCAM Explainability

[GitHub](#)

PyTorch · CNN · GradCAM · GradCAM++ · Data Augmentation

- 91% accuracy** on **4,200+ MRI scans** across **4** tumour categories with GradCAM heatmap explainability.
- Interpretable predictions support clinical review workflows.
- Designed hybrid **EfficientNet-B0** + **Swin Transformer** architecture with attention-based feature fusion for high-fidelity classification.

4. AIMLverse — Interactive ML Education Platform

[GitHub](#)

Next.js 14 · Three.js · Python · Interactive Visualisation

- Production-deployed platform at aimlverse.in with **3D visualisers** for LLM, CNN, and MLP architectures.
- Live backpropagation, normalisation, and decision boundary demos built from scratch using raw Three.js and custom Python backends.

5. Face Cursor Control via Facial Landmark Detection

[GitHub](#)

MediaPipe · OpenCV · Python Multithreading · Real-time CV

- Developed during a 2nd-year hackathon for accessibility; used multithreading to achieve **40% speedup**, enabling webcam-based nose cursor control and wink-triggered clicks without additional hardware.